

# Cora “Chanel” Richardson

Data strategist with a background in risk governance, data engineering, and computational redistricting. Proven track record in developing technical solutions for civil rights litigation and architecting responsible AI frameworks within global-scale cloud environments. Expert in translating technical risks for legal and executive stakeholders.

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## Technical Skills

- **NLP & ML:** LLMs/SLMs (Vertex AI, Transformers), Text Representation (TF-IDF, BERT, Embeddings), Fuzzy Matching, Cosine Similarity, Sentiment Analysis
- **Programming:** Python (Flask, NLTK, BeautifulSoup, Pandas, Geopandas, SciKitLearn, Keras, Tensorflow), Rust, SQL, Javascript / Typescript, C++, HTML
- **Data Ops & Cloud:** Google Cloud Platform (GCP), BigQuery, AWS (EC2 & Lambda), Apache Airflow, Docker, CI/CD, Git

## Professional Experience

### Google, Data Engineering & Modeling Lead, Cloud Risk & Compliance (Oct 2025 – Present)

- Lead end-to-end data strategy for Google Cloud’s Risk & Compliance org, architecting scalable platforms using Python, SQL, and TypeScript on GCP to streamline data access for cross-functional stakeholders.
- Direct "Project Constellation," an advanced modeling initiative to predict concentrated cybersecurity and privacy risks; translated complex statistical outputs into actionable technical roadmaps for product adoption.
- Architected a unified data ecosystem by consolidating seven years of siloed audit and compliance data through custom APIs and a centralized UI, improving data reliability and organizational visibility.
- Serve as a primary technical-to-strategic translator, bridging the gap between engineering teams and legal leads to ensure technical development aligns with evolving regulatory mandates and high-stakes strategy.
- Engineered novel AI applications comparing SLMs and LLMs, developing rigorous frameworks to evaluate model performance and surface non-obvious insights in high-uncertainty environments.
- Developed custom diagnostics to analyze and compare AI model performance for risk reporting, ensuring accuracy and catching anomalies in complex, messy datasets.

### Google, Data Analyst, Cloud Risk Reporting (June 2023 – Oct 2025)

- Spearheaded "Risk Reporting 2.0," transitioning the organization from reactive reporting to a proactive, historical analysis methodology used to identify and mitigate emerging risks.
- Architected suite of Looker dashboards centralizing Key Risk Indicators (KRIs) across multiple domains, providing leadership with real-time visibility into the organization’s risk posture.
- Orchestrated cross-functional metric onboarding, collaborating with diverse teams to define, validate, and visualize high-impact data points for executive-level risk reports.
- Designed a multi-stage data flow integrating scraped content with Vertex AI (LLM) for automated insight generation.

### Metric Geometry and Gerrymandering Group, Research Analyst (August 2021 - May 2023)

- Co-authored and maintained the [gerrytools](#) Python package, an open-source library used by the redistricting community and legal teams, contributing to internal GitHub workflows and external documentation.
- Analyzed large voter files from Texas and North Carolina to perform ecological inference simulations, determining the statistically probable voting outcomes of districts across each state to analyze potential partisan or racial gerrymandering.
- Developed and executed complex computational and statistical analyses using GIS, Python, and Rust on massive geospatial datasets used to support litigation across eight states, ensuring data integrity and reliability for high-stakes legal proceedings.
- Interfaced with expert witnesses and lawyers to translate technical findings into clear narratives, demonstrating strong stakeholder communication and commitment to clear process documentation.

## Education

### Duke University

M.S. Electrical Engineering

2020-2021

### Tufts University

B.S. Computer Engineering

2016-2020

## Talks & Publications

- "Using Python to Protect Voting Rights," SciPy Conference, July 2022
- "Graphs, Stats, and Math Take the Stand: The Role of Mathematics in Redistricting Litigation," Math Society Conference, Brown University, October 2022
- "Ecological Inference & Voting Rights," Joint Mathematics Meetings, January 2023
- "Aggregating Community Maps," SIGSPATIAL '22: Proceedings of the 30th International Conference on Advances in Geographic Information Systems (co-authored)